Algorithms and Computability

Lecture 3: Computability

Martin Zimmermann (Aalborg University)

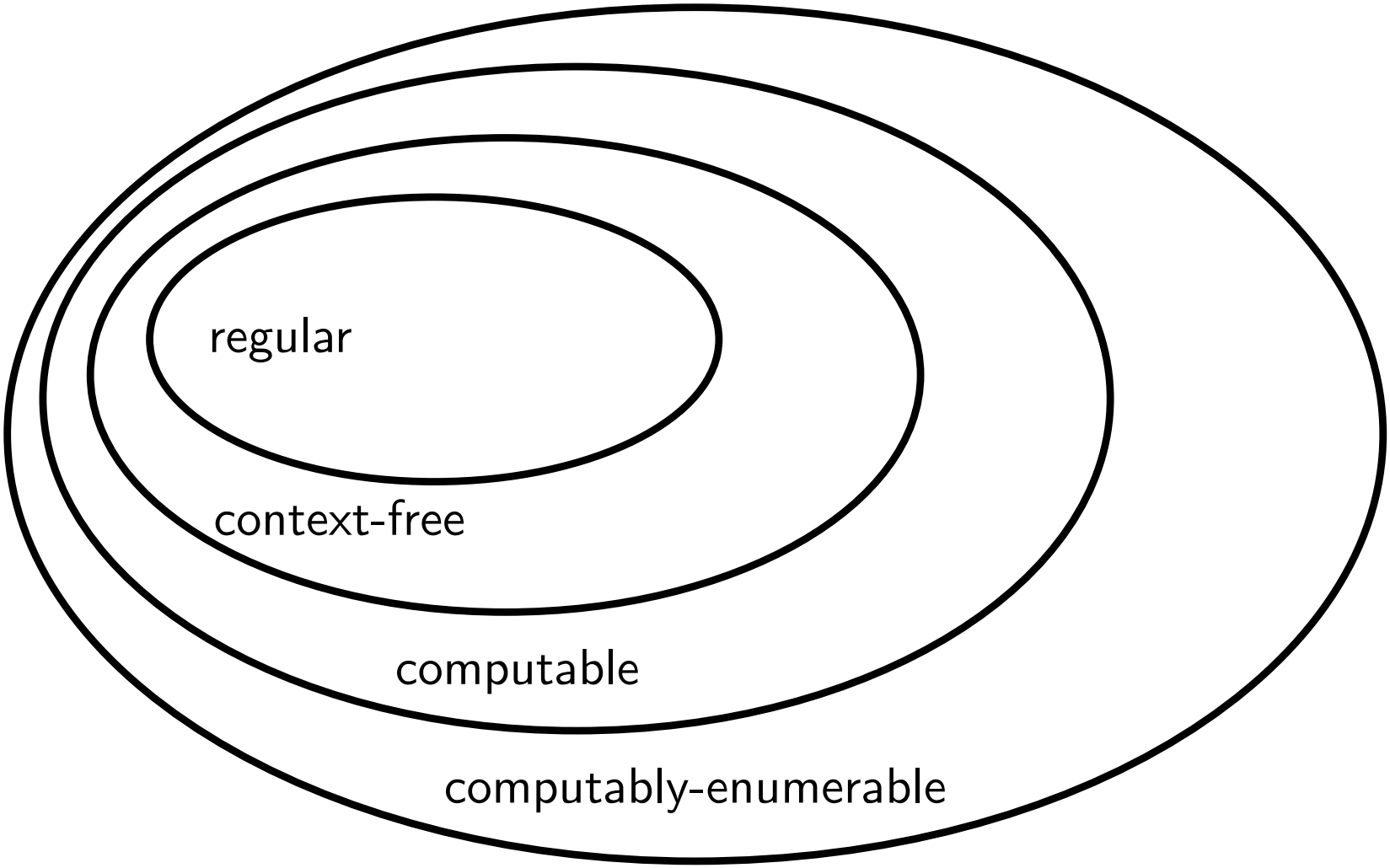
Last Week in Algorithms and Computability

We have seen

- the Church-Turing Thesis ("Everything that can be computed can be computed by a Turing machine"),
- multi-tape DTM's and
- nondeterministic TM's, and
- their equivalence: The same classes of languages are computably-enumerable (computable) by
  1. Deterministic one-tape TM's,
  2. Deterministic multi-tape TM's,
  3. Nondeterministic one-tape TM's, and
  4. Nondeterministic multi-tape TM's.

Today

Understand the structure of computable and computably-enumerable languages:



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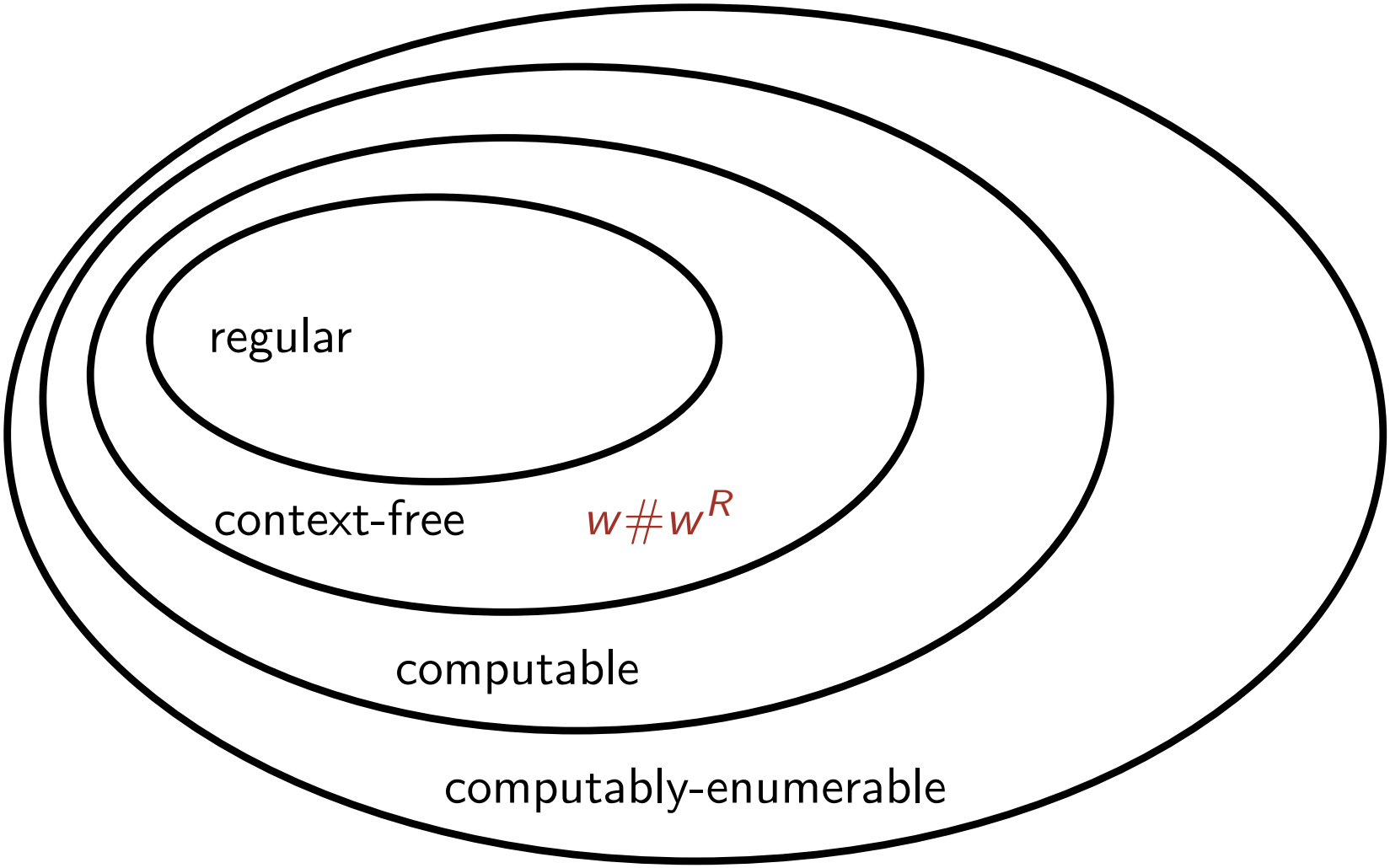

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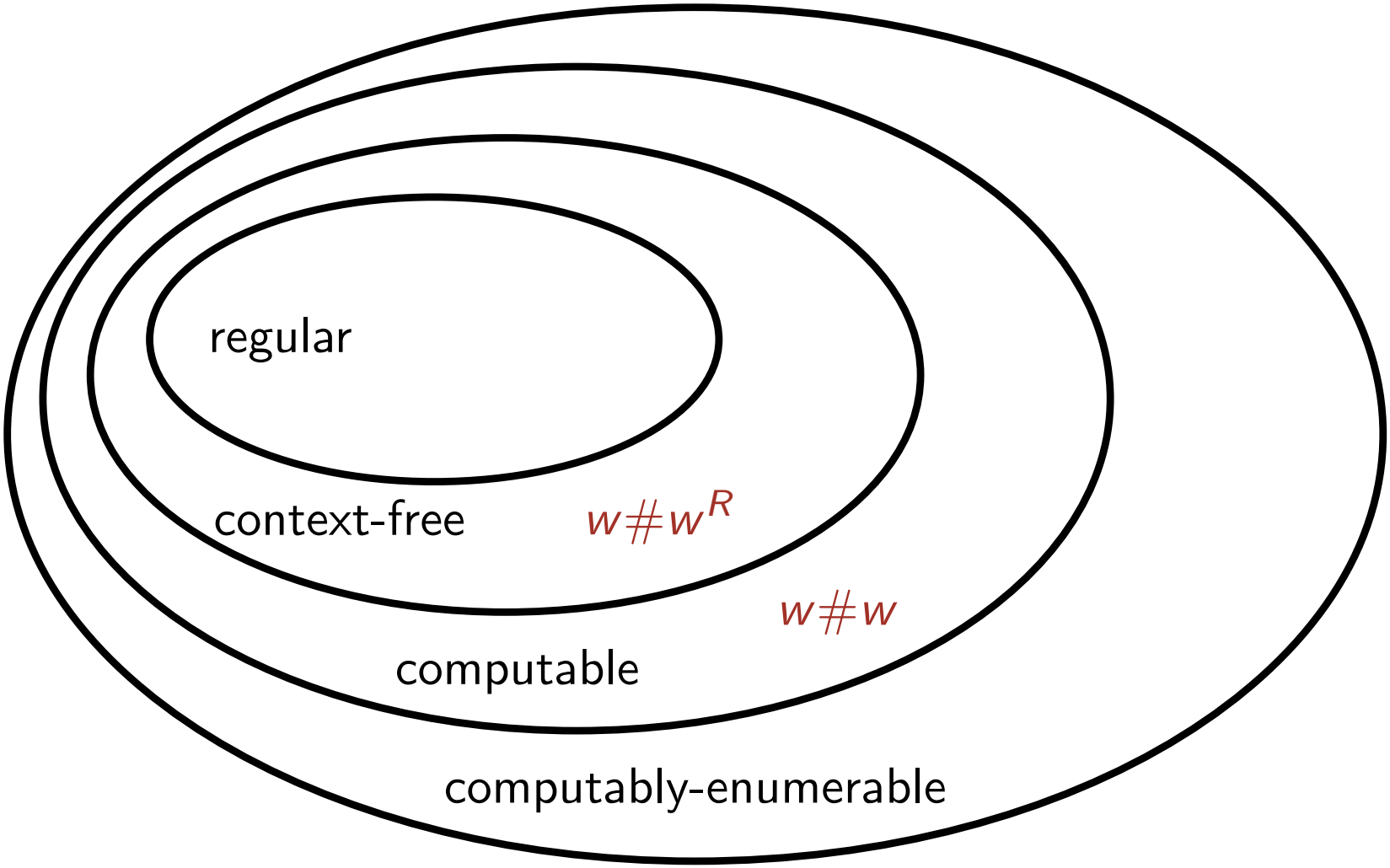

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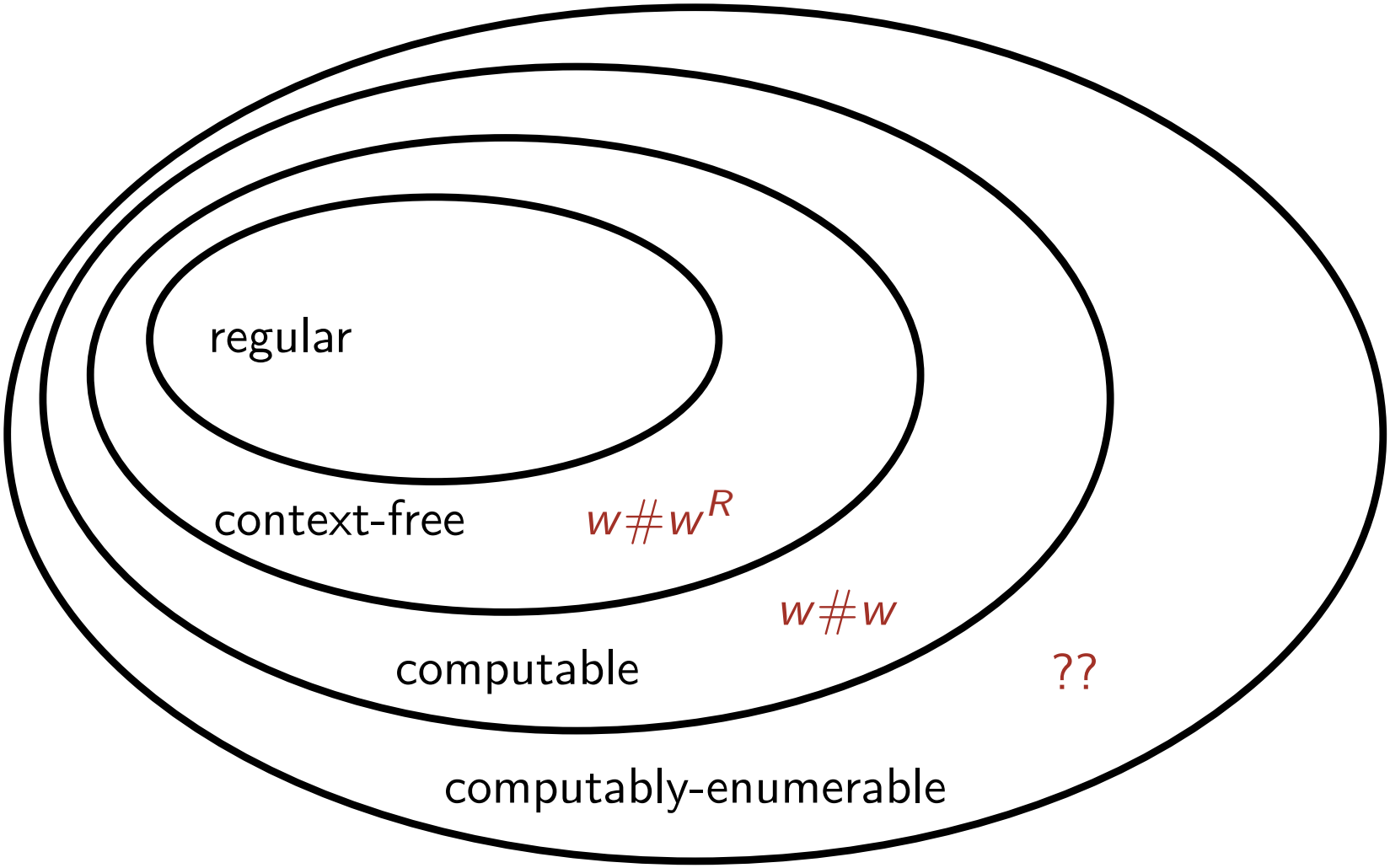

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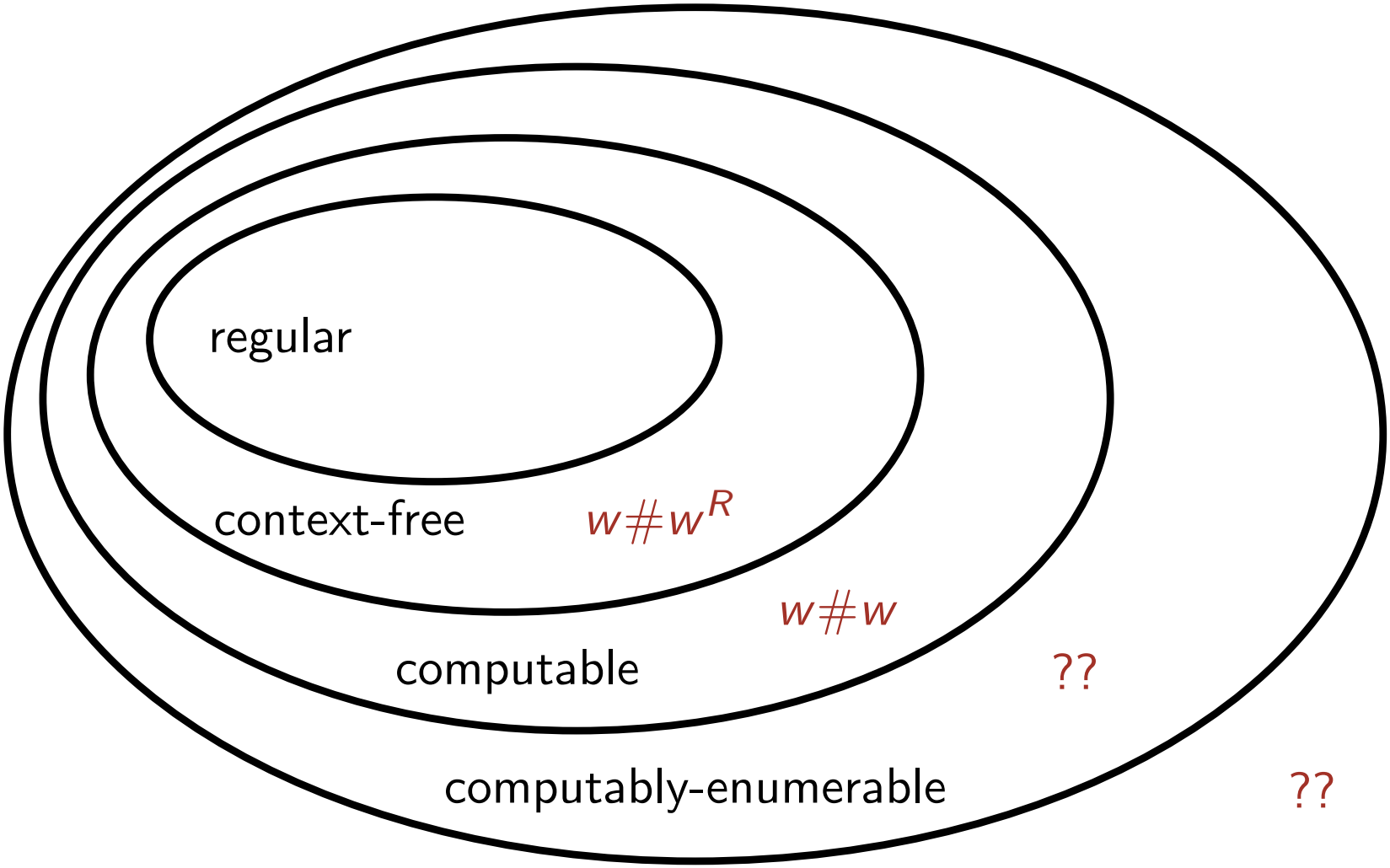

Agenda

1. **Warm-up: Hotels and Barbers**
2. **The Halting Problem**
3. **Closure Properties**

Hilbert's Grand Hotel

You have a hotel with ten rooms, all occupied. Can you accommodate a newly arriving guest?

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Infinite sets (and infinite hotels) behave quite differently than finite ones!

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- For finite sets: if they have the same (finite) number of elements.

$$\left\{ \begin{array}{c} 3, \\ 7, \\ 13 \end{array} \right\} \qquad \left\{ \begin{array}{c} a, \\ p, \\ u \end{array} \right\}$$

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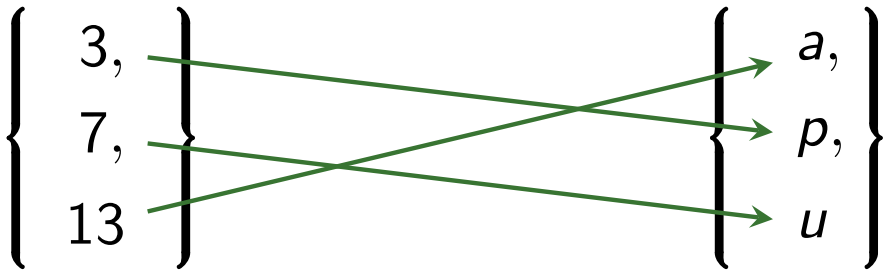

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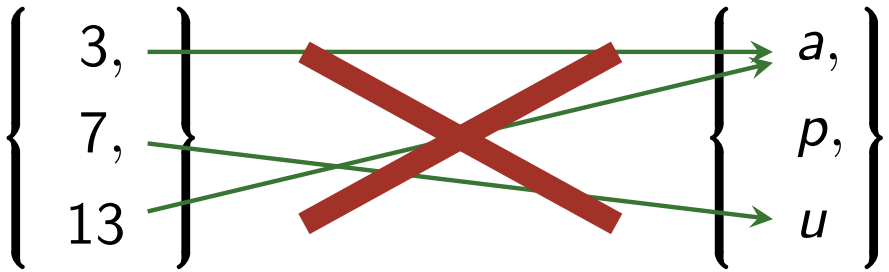

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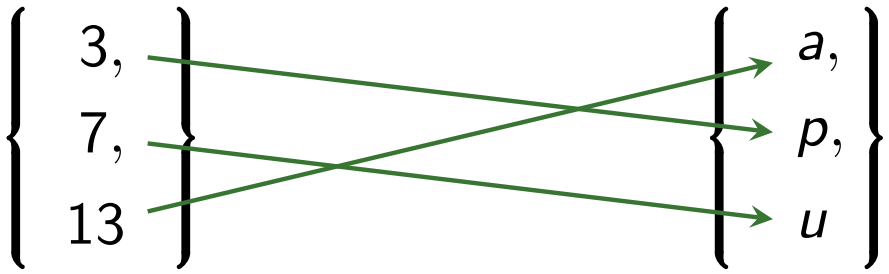

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We use the same approach for infinite sets.

Unintuitive Consequences

"Two sets have the same cardinality if there is a bijection between them."

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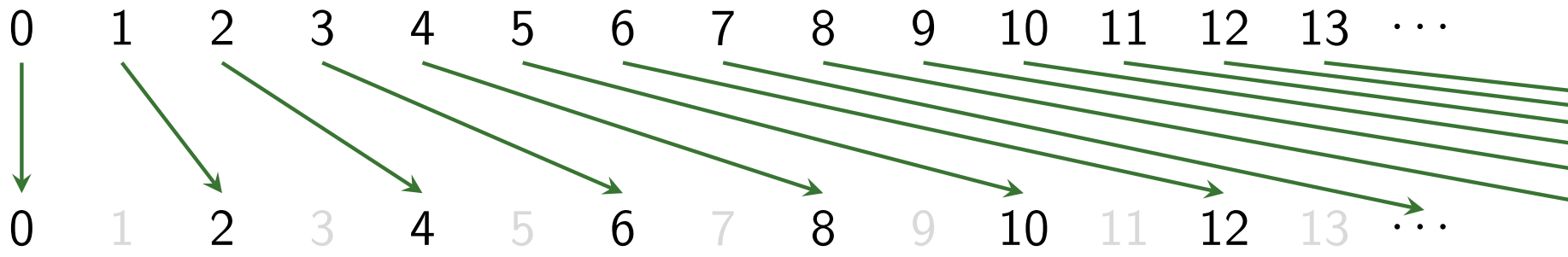

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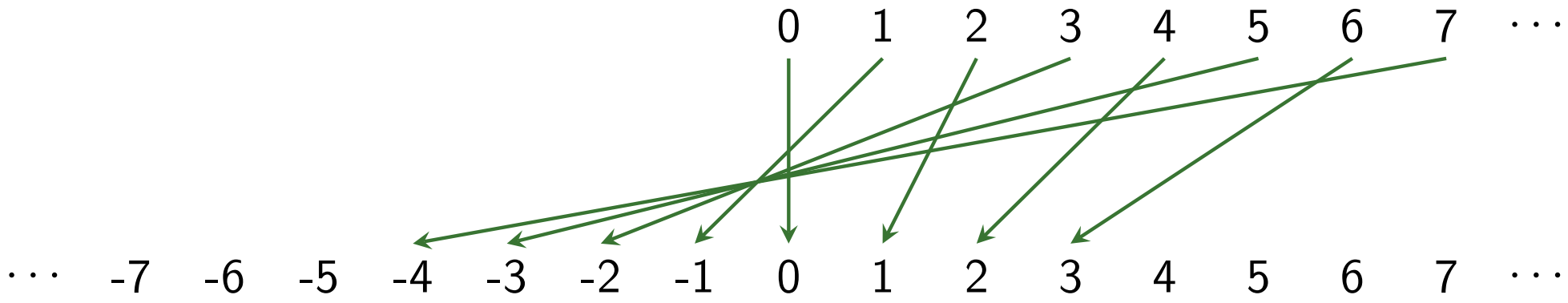
- So, the natural numbers and the even natural numbers have the same cardinality.
- And the natural numbers and the integers have the same cardinality.



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- And the natural numbers and the integers have the same cardinality.
- The natural numbers and Σ^* have the same cardinality.
- The latter result also implies that even the natural numbers and the rational numbers have the same cardinality.
- But using techniques shown on the next slide show that the cardinality of the reals is strictly larger than that of the natural numbers.

Back to our Question

- Assume the natural numbers and the subsets of the natural numbers have the same cardinality.
- Then, there is a bijection f between the natural numbers and the subsets of the natural numbers.
- So, the list $f(0), f(1), f(2), \ldots$ contains all subsets of the natural numbers.

Back to our Question

- We can arrange the list $f(0), f(1), f(2), \ldots$ in an infinite table such that row i represents the subset $f(i)$.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	$\cdots$
$f(0)$	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	$\cdots$
$f(1)$	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	$\cdots$
$f(2)$	✗	✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✓	✗	✓	$\cdots$
$f(3)$	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	$\cdots$
$f(4)$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	$\cdots$
$f(5)$	✗	✓	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	$\cdots$
$f(6)$	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	$\cdots$
$f(7)$	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

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- We can arrange the list $f(0), f(1), f(2), \ldots$ in an infinite table such that row i represents the subset $f(i)$.
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- It differs from every set $f(i)$ in the membership of i.

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- This is a contradiction to our assumption that the list contains all subsets, i.e., our assumption must be wrong: there is no bijection between the natural numbers and the subsets of natural numbers.

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This process, due to Georg Cantor, is called **diagonalization** and has many applications in set theory, logic, and computability.

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Can we apply self-reference to Turing machines?

Agenda

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2. **The Halting Problem**
3. Closure Properties

Encoding of Turing Machines

Let $M = (Q, \Sigma, \Gamma, s, t, r, \delta)$ be a DTM. We assume without loss of generality $Q = \{1, 11, \ldots, 1^{|Q|}\}$ and $\Sigma = \{0, 1\}$. Also, let $rep_\Gamma \colon \Gamma \to \{1, 11, \ldots, 1^{|\Gamma|}\}$ be an encoding of Γ such that $rep_\Gamma(0) = 1$, $rep_\Gamma(1) = 11$, and $rep_\Gamma(\sqcup) = 111$.

Encoding of Turing Machines

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Then, M is encoded by the word $\ulcorner M \urcorner$ over $\{0, 1\}$ defined as follows:

$$1^{|Q|}\, 0\, 1^{|\Gamma|}\, 0\, s\, 0\, t\, 0\, r\, 0\, w_\delta$$

where w_δ is the list of encodings of transitions. Each $\delta(q, b) = (p, a, d)$ is encoded by

$$q\, 0\, rep_\Gamma(b)\, 0\, p\, 0\, rep_\Gamma(a)\, 0\, dir(d)\, 0$$

where $dir(-1) = 1$ and $dir(+1) = 11$.

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Note

Basis of modern computer architecture: program and input are both data stored in memory (von Neumann architecture).

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- Here, $\langle \cdot, \cdot \rangle$ is a function that encodes two words x, y over $\{0, 1\}$ by a single word $\langle x, y \rangle$ over $\{0, 1\}$. See the "Computability and Complexity" (page 89) for details.

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Theorem (Turing 1936)

The halting problem is not computable.

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- List behaviour of all halting DTM's M_i on all inputs w_j in matrix.

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	w_0	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	w_9	$\cdots$
M_0	acc	rej	acc	rej	acc	rej	acc	rej	acc	rej	$\cdots$
M_1	rej	acc	rej	acc	rej	acc	rej	acc	rej	acc	$\cdots$
M_2	rej	rej	acc	acc	rej	acc	rej	acc	rej	rej	$\cdots$
M_3	rej	rej	rej	rej	rej	rej	rej	rej	rej	rej	$\cdots$
M_4	acc	acc	acc	acc	acc	acc	acc	acc	rej	acc	$\cdots$
M_5	rej	acc	acc	rej	acc	rej	rej	rej	acc	rej	$\cdots$
M_6	acc	rej	rej	acc	rej	rej	acc	rej	rej	acc	$\cdots$
M_7	acc	rej	rej	acc	rej	rej	rej	rej	rej	acc	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Intuition

- List behaviour of all halting DTM's M_i on all inputs w_j in matrix.
- Flipping the diagonal gives language $\{w_3, w_5, w_7, \ldots\}$ that is different from every computable language.

	w_0	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	w_9	$\cdots$
M_0	acc	rej	acc	rej	acc	rej	acc	rej	acc	rej	$\cdots$
M_1	rej	acc	rej	acc	rej	acc	rej	acc	rej	acc	$\cdots$
M_2	rej	rej	acc	acc	rej	acc	rej	acc	rej	rej	$\cdots$
M_3	rej	rej	rej	rej	rej	rej	rej	rej	rej	rej	$\cdots$
M_4	acc	acc	acc	acc	acc	acc	acc	acc	rej	acc	$\cdots$
M_5	rej	acc	acc	rej	acc	rej	rej	rej	acc	rej	$\cdots$
M_6	acc	rej	rej	acc	rej	rej	acc	rej	rej	acc	$\cdots$
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- Assume the halting problem is computable, i.e., there is a halting DTM H for the halting problem.

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- Case 1: B halts on input $\ulcorner B \urcorner$. That means H accepts $\langle \ulcorner B \urcorner, \ulcorner B \urcorner \rangle$ by definition of the halting problem. Hence, B does not halt on input $\ulcorner B \urcorner$.

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- Case 2: B does not halt on input $\ulcorner B \urcorner$. That means H rejects $\langle \ulcorner B \urcorner, \ulcorner B \urcorner \rangle$ by definition of the halting problem. Hence, B halts on input $\ulcorner B \urcorner$.

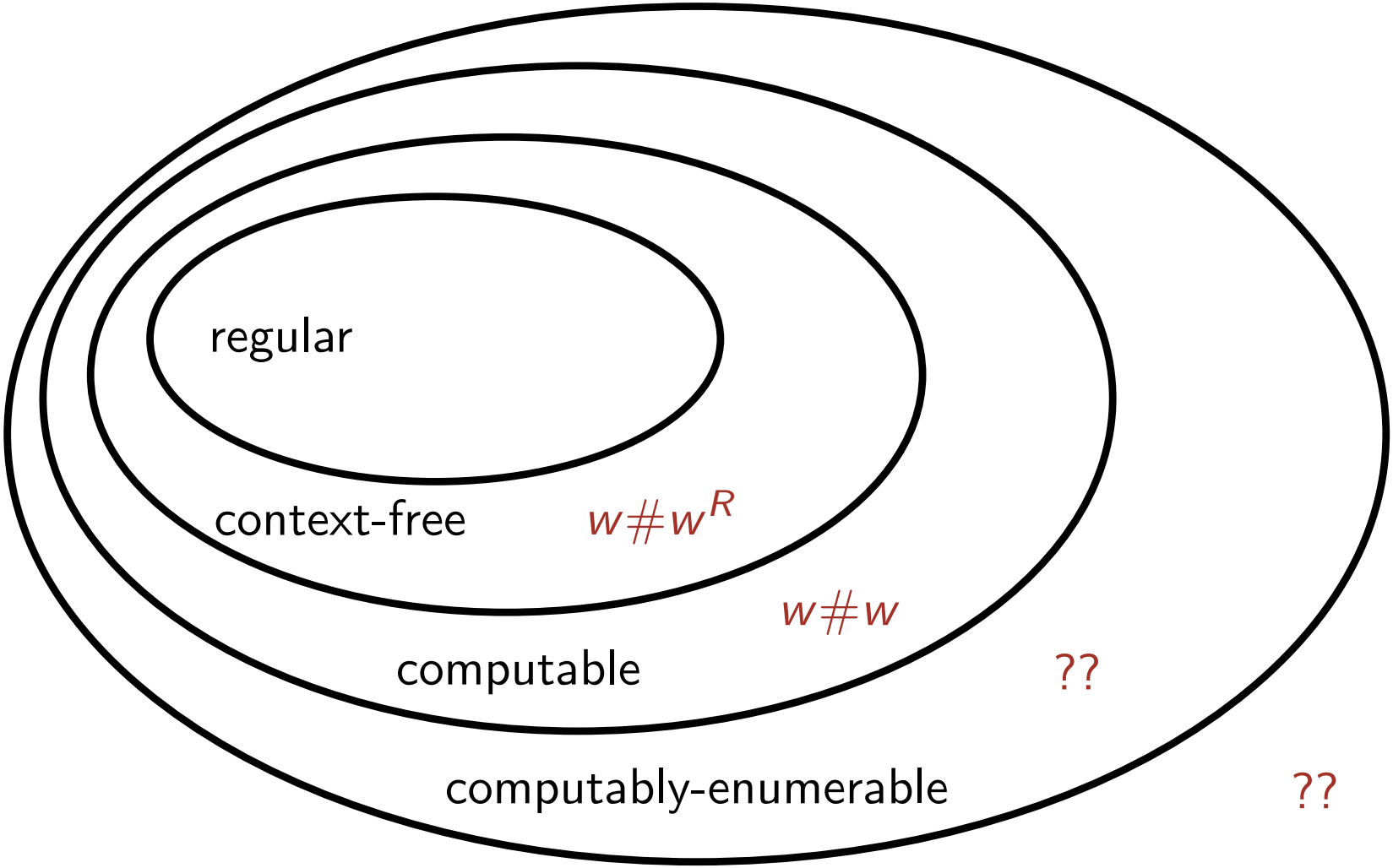
Both cases lead to a contradiction, so H cannot exist. Hence, the halting problem is not computable.

What about Computably-enumerability?

Is the halting problem at least computably-enumerable?

Exercise 4 this week: Yes, it is!

What Have We Achieved?



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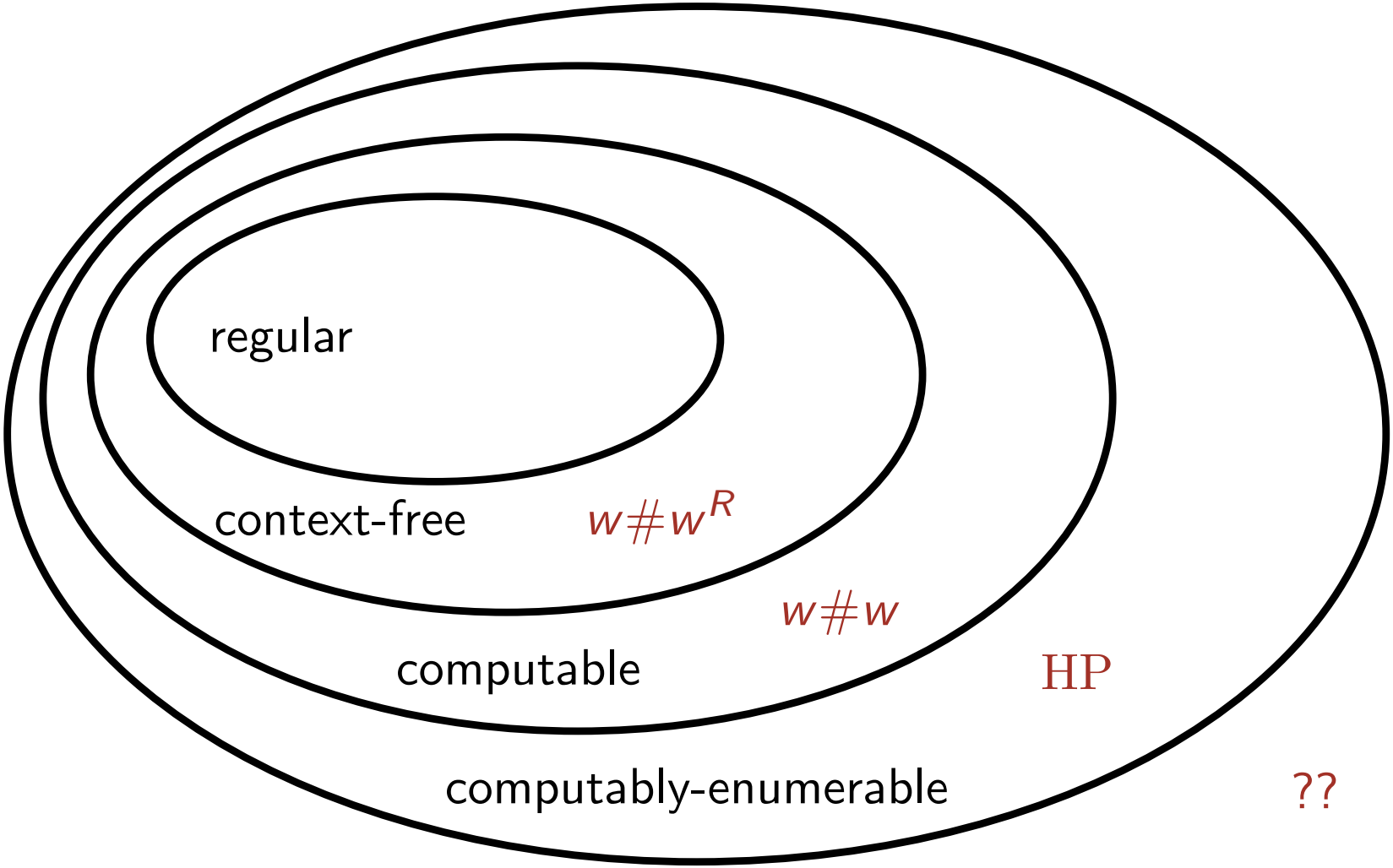

Agenda

1. Warm-up: Hotels and Barbers
2. The Halting Problem
3. **Closure Properties**

Reminder: Operations on Languages

Let L_1, L_2 be two languages over Σ.

- union:

$$L_1 \cup L_2 = \{w \in \Sigma^* \mid w \in L_1 \text{ or } w \in L_2\}$$

- intersection:

$$L_1 \cap L_2 = \{w \in \Sigma^* \mid w \in L_1 \text{ and } w \in L_2\}$$

- complement (w.r.t. Σ^*):

$$\overline{L_1} = \{w \in \Sigma^* \mid w \notin L_1\}$$

- concatenation:

$$L_1 \cdot L_2 = \{w \in \Sigma^* \mid w = w_1 w_2 \text{ with } w_1 \in L_1 \text{ and } w_2 \in L_2\}$$

- Kleene star (iteration):

$$(L_1)^* = \{w \in \Sigma^* \mid w = w_1 w_2 \cdots w_k \text{ for some } k \geq 0 \text{ and } w_i \in L_1 \text{ for all } i \in \{1, 2, \ldots, k\}\}$$

Why Care about Closure Properties?

There are two main applications of closure properties:

1. Show that a language is in some class of languages.

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Example

- Regular languages are closed under intersection.
- $L_s = \{w \in \mathbb{B}^* \mid w \text{ starts with a } 1\}$ is regular.
- $L_e = \{w \in \mathbb{B}^* \mid w \text{ ends with a } 1\}$ is regular.
- So, $\{w \in \mathbb{B}^* \mid w \text{ starts and ends with a } 1\} = L_s \cap L_e$ is regular as well.

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- Assume $L_{ne} = \{x\#y \mid x, y \in \mathbb{B}^* \text{ such that } x \neq y\}$ is regular.
- Then, $L = \overline{L_{ne}} \cap L_{prs}$ is regular as well.
- But, $L = \{w\#w \mid w \in \mathbb{B}^*\}$, which we know is not regular.
- Contradiction. So, our assumption is wrong and L_{ne} is not regular.

Intersections

Given one-tape DTM's M_1 for $L_1 \subseteq \Sigma^*$ and M_2 for $L_2 \subseteq \Sigma^*$, we construct a two-tape DTM for $L_1 \cap L_2$:

On input w:

1. Copy w on second tape, place both heads at first letter of w.
2. Do one step of M_1 on first tape and one step of M_2 on second tape (if one halts only simulate the other).
3. If both M_1 and M_2 accept, then accept as well.
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On input w:

1. Nondeterministically choose $i \in \{1, 2\}$.
2. Run M_i on w.
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Given a deterministic one-tape **halting** DTM M for $L \subseteq \Sigma^*$, we construct a deterministic halting DTM for $\overline{L}$:

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Does the same construction work for computably-enumerable languages?

Closure Properties

Theorem

1. *The computable languages are closed under union, intersection, complement, concatenation, and iteration.*
2. *The computably-enumerable languages are closed under union, intersection, concatenation, and iteration, but* **not** *under complement.*

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- We sketched some of the proofs here, but not all.
- Some more are on this week's exercise sheet.
- The others are similar (and can be found in the literature).

Conclusion

We have seen

- Diagonalization
- and the non-computability of the halting problem.
- Also: closure properties of computable and computably-enumerable languages.

Reading

None today, wait till tomorrow.